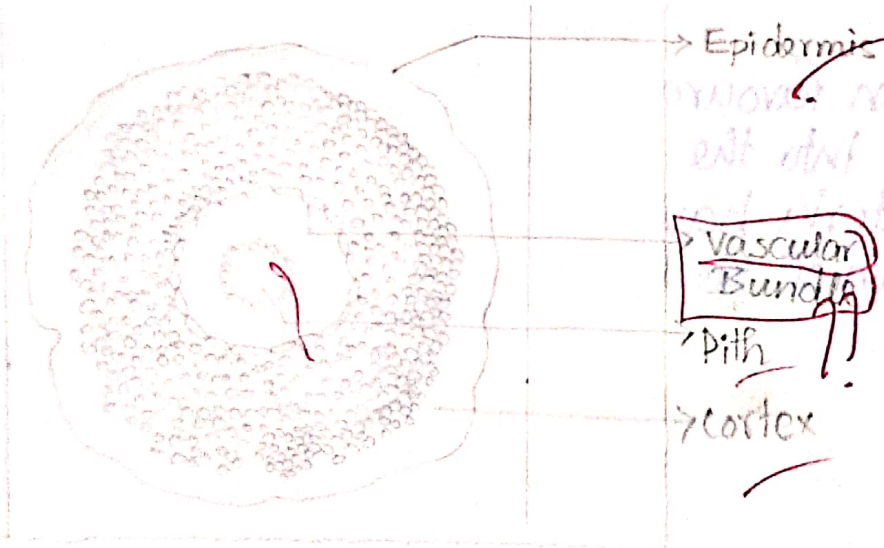


[Fig- T.S of root parasite Orobranchae sp.] (microscopic view in photoplate)



[Fig→ T.S of root parasite Orobranche sp.]
(microscopic view labelled diagram)

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Study of Biotic Interaction of Root Parasite :-

Root parasite is a plant that taken nutrients from host plant by attaching and to penetrating to the root system. Root parasites penetrates the host root system and connect to its vascular system by a specialized structure called haustoria. The parasite of the family orobranchae (orobranche sp.) attach to the roots of their host and draw nutrients from the vascular tissue. This parasite grow primarly on roots of sunflower, tomato and leguminous plants. Orobranchae sp. are obligate parasite. i.e- they must attach through host plants root to survive.

Morphology :-

(i) This parasite plants enters the host plant through its root system.
(ii) The colour of the parasitic plant ranges from white to yellowish -brown.
(iii) Lack of photosynthetic tissue.
(iv) The haustorium develope from roots is embeded inside of host and serving to feed the parasite located externally to host.
(v) The haustorium develop tuberous nodule at the attachment side of host.

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Anatomy:

A transverse section through both plants where the root is attached to its host, the parasite root is fixed to the host root by a dice shaped tubular haustorium that has grown into the host root intercellulary upto the vascular bundle. Each haustorium has a central xylem continuous with the vascular bundle of the host root.

Apart from this orobranchae have a vascular system with a ring of vascular bundle is composed of phloem cambium and xylem. The xylem and phloem with this bundles are responsible for uptake the nutrients from the host to a parasite. The cortex of the roots contains parenchyma cells which are involve in storage and metabolic process.

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